

Fenrix Strategy Validation Lab

Backtesting shows where a strategy worked. Fenrix searches for where it breaks.

- Product flow: Describe → Review → Lock → Backtest → Stress → Minimize → Export
- Strategy hash (immutable): c9d19b2e8d20667a...
- Source code SHA: 83bbe5f262d7bbac0f4ce43890dc8e064f8b7a90 · data mode of record: yfinance · universe: 30 assets

The problem

A green backtest is a claim, not a validation

- Backtests only tell you where a strategy already worked — they are silent about fragility.
- Students and retail quants ship strategies validated on one historical path.
- Overfitting, cost blindness and regime fragility hide until real capital finds them.
- There is no cheap, reproducible way to ask: under which conditions does this break?

Product workflow

Describe → Review → Lock → Backtest → Stress → Minimize → Export

- Describe: plain-English strategy → structured clause ledger (every clause reviewed).
- Review + Lock: mandatory approve step → immutable version + canonical SHA-256 hash.
- Backtest: the SAME locked hash runs the historical backtest on the run-of-record data (yfinance).
- Stress: the SAME hash enters a sealed synthetic failure search across mechanisms.
- Minimize: confirmed failures are shrunk to a minimal reproducible counterexample.
- Export: evidence package with manifest, hashes and claim ledger.

← Strategy Lab

Fenrix Submission MVP

source: —

git: —

hash: —

One story, eight verifiable stages: plain-English strategy → reviewed clause ledger → mandatory approval → immutable hash → declared data source → real multi-asset backtest → sealed synthetic stress → minimized failure + adjacent pass → signed evidence.

▼ 🌟 New here? How to use this page

1. Pick a universe — the list of tickers your strategy trades. For SAA/TAA class work, choose “SAA / TAA building blocks” (asset-class ETFs: stocks, bonds, gold, commodities...), or pick **Custom** and type your own comma-separated tickers.

2. Optionally pick a strategy goal — a plain-English starting point (60/40, risk parity, momentum). Note: these are goal descriptions; the engine always runs the flagship long/short momentum model. Its description appears below the controls.

3. Click “▶ Run live pipeline” to compile, approve, backtest and stress-test your setup end-to-end. Or click “✓ Run verified judge demo” to see a finished, canned example first.

4. Read the 8 stages as they light up, then download the signed evidence JSON at the end (stage 8).

Everything here is for research and education only — it is not investment advice.

Universe preset

Flagship 29 (default)

▼

Example strategy goal — not executed

Flagship L/S momentum (default — the engine that actually runs)

▼

Templates are plain-English starting points only. Every run executes the flagship long/short momentum engine, the SAA/TAA labels describe a goal, not a different allocator. Full SAA/TAA execution is on the roadmap.

Mode

Tier 3 · deterministic synthetic (offline)

▼

Stress budget

8

▶ Run live pipeline

✓ Run verified judge demo

Flagship L/S momentum (default) · universe: Flagship 29 (default) — 30 tickers

Fenrix flagship long/short momentum-volatility strategy; rank the universe cross-sectionally by momentum and volatility, hold the top ranks long and bottom ranks short, rebalance on schedule with cost controls. This is what the pipeline executes.

● 1 · Strategy

● 2 · Clause review

● 3 · Approval

● 4 · Data source

● 5 · Historical results

● 6 · Sealed stress

● 7 · Failure replay

● 8 · Evidence export

Live log

idle...

Charts are hand-rolled inline SVG (zero external dependencies). The verified judge demo renders persisted, deterministic evidence with no live-server dependency. Synthetic stress worlds probe fragility, not real future risk; costs are labeled heuristics, not broker-calibrated.

Demo run — Real historical backtest (yfinance)

30 assets · 50 trades · all figures from deck_data.json

13.07%

cumulative return

0.18

Sharpe

-26.75%

max drawdown

14.65%

ann. volatility

14.03%

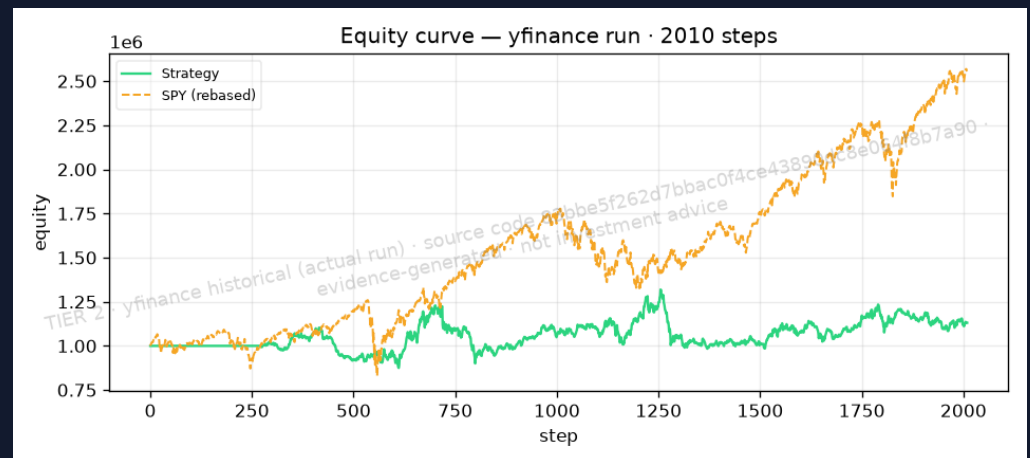
cost % of capital

50

trades

- CAGR 1.55% · Sortino 0.22 · Calmar 0.06
- Benchmark CAGR 12.49% · information ratio -0.46
- Gross exposure avg 103.70% · net exposure avg 0.00%
- Costs are explicit bounded heuristics (commission/spread/slippage/borrow) — not broker calibrations.

TIER 2



What the normal backtest missed

Sealed stress search — per-mechanism pass/fail (base seed)

- Baseline (Real historical backtest (yfinance)): Sharpe 0.18, max drawdown -26.75%, cumulative return 13.07%.
- Sealed stress search: 8 confirmed failures out of 24 worlds (33.33% failure rate).
- Confirmed-failure mechanisms: borrow_cost_increase, correlation_breakdown, slippage_inflation, volatility_expansion.
- Worst confirmed stress case (volatility_expansion): Sharpe 0.20, max drawdown -0.66%, cost 0.10% of capital — invisible to the plain backtest.

Per-mechanism result (base seed):

- momentum_reversal: pass
- volatility_expansion: FAIL (low_sharpe)
- correlation_breakdown: FAIL (low_sharpe)
- volatility_compression: pass
- spread_inflation: pass
- slippage_inflation: FAIL (low_sharpe)
- borrow_cost_increase: FAIL (low_sharpe)
- short_unavailability: pass
- delayed_rebalance: pass
- universe_churn: pass
- missing_data_shock: pass

TIER 2

Minimized counterexample

Smallest reproducible world where the locked strategy still fails

- Baseline vs minimized failure — the breaking condition, shrunk to a minimal reproducible counterexample:
- • BASELINE (Real historical backtest (yfinance)): Sharpe 0.18, cum. return 13.07%.
- • MINIMIZED FAILING WORLD (volatility_expansion @ intensity 0.9285, seed 20264): still fails = True; violated predicates = low_sharpe.
- minimized-case metrics: Sharpe 0.20, cum. return 0.69%, max DD -0.66%, cost 0.10% of capital.
- • Adjacent pass: no adjacent pass within radius — reported honestly, not fabricated.

TIER 2

Education & research value

A falsification lab, not a profit promise

- Students see WHY a strategy fails — mechanism, intensity, seed — not just an equity curve.
- Every claim is bound to a hash and an evidence file; grading and review become reproducible.
- Instructors can replay the minimized counterexample deterministically.
- Mechanism library searched this run: borrow_cost_increase, correlation_breakdown, delayed_rebalance, missing_data_shock, momentum_reversal, short_unavailability, slippage_inflation, spread_inflation, universe_churn, volatility_compression, volatility_expansion.

Architecture & moat

Immutable contracts + sealed synthetic worlds

- Clause ledger → canonical hash: the same immutable artifact flows through every stage.
- Sealed synthetic world generator: seeds + mechanisms are reproducible and tamper-evident.
- Evidence packages carry SHA-256 manifests binding data, backtest, campaign and replay.
- Failure minimizer + adjacent-pass search turn red flags into actionable counterexamples.
- Moat: the corpus of confirmed, minimized failures compounds with every run.

Honest boundary & roadmap

What this is NOT — and what comes next

- Synthetic stress worlds are generated, not historical; they probe fragility, not real future risk.
- Costs are explicit bounded heuristics (flat bps), not validated broker calibrations.
- Fenrix fundamentals (if used) are not point-in-time; lagged approximation only.
- No claim of live profitability, production execution fidelity, or universal realism.
- Failure search is not exhaustive; it reports confirmed failures only.
- Roadmap: point-in-time fundamentals, broker-calibrated costs, richer mechanism library, live paper-trading bridge.

Ask

Help us make strategy falsification the default

- Pilot with quant-finance classrooms and student funds this semester.
- Feedback on mechanism realism from practitioners.
- Compute credits for larger sealed stress campaigns.
- Reproduce everything: make submission-demo · make verify-submission · make pitch-deck.